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Module 5 Introduction

Lesson 1: Parameterized Priors and the Concept of Exchangeability

Lesson 2: Analysis and Applications of Hierarchical Models

Reading: Analysis and Applications of Hierarchical Models Readings
3h

Video: Hierarchical Models Pt. 1
3 min

Video: Hierarchical Models Pt. 2
3 min

Video: Hierarchical Models Pt. 3
4 min

Video: Hierarchical Models Pt. 4
7 min

Practice Assignment: Analysis and Applications of Hierarchical Models Quiz
Submitted

Lesson 3: Computational Techniques and Model Validation

Module 5 Summative Assessment

Module 5 Summary

Your grade: 100%

Next item

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Analysis and Applications of Hierarchical Models Quiz

Instructions

Review Learning Objectives

1. What is exchangeability in the context of hierarchical models? **1 / 1 point**

- Indexes aren't meaningful.
- Parameters follow a uniform distribution.
- Parameters are equal.
- Parameters are independent.

Coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Correct
Correct! Exchangeability means the joint distribution of parameters is invariant to permutations of their indexes.

Assignment details
Submitted Dec 6, 1:08 AM IST
Attempts Unlimited
Retry

2. How can exchangeability be affected when additional information about units is available? **1 / 1 point**

- Exchangeability is replaced by partial exchangeability.
- Exchangeability is enhanced.
- Exchangeability may be partial or conditional.
- Exchangeability is lost.

Your grade
100%
View submission See feedback

Correct Like Dislike Report an issue
Additional information can lead to partial or conditional exchangeability, where the exchangeability holds given certain conditions or within subsets of the data.

3. Why is it important to check the integrability of the posterior distribution when using improper priors? **1 / 1 point**

- To verify that the resulting posterior is proper.
- To ensure computational efficiency.
- To avoid overfitting the data.
- To simplify the prior distribution and make simulation possible.

Correct
Correct! It is crucial to ensure the resulting posterior distribution is proper.